SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

MTD215

Timed Online Assignment - July Semester 2022

Application of C++ in Multimedia

Tuesday, 22 November 2022 **01:00 pm - 03:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **FOUR (4)** questions and comprises **EIGHT (8)** pages (including cover page).

2. You must answer **FOUR (4)** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a Microsoft Word file saved in .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the Word file provided it does not exceed the file size limit of 500MB. Images of handwritten answers will not be marked.

6. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

7. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

MTD215 Copyright © 2022 Singapore University of Social Sciences (SUSS)

(Full marks: 100)

Section A (100 marks)
Answer all questions in this section.

Question 1

Implement a C++ program and explain the syntax and use of data structures such as structures and class.

Question 1a

Define a structure named Person using the C++ language to hold

- A string representing the name of the person.
- A float representing the BMI of the person.
- An integer representing the age of the person.
- A bool representing the Singapore nationality status of the person.

(5 marks)

Question 1b

Implement a main function to demonstrates the use of the structure defined in question 1a. The function declares an array of ten person items. Define the last element in the array as follows:

name : "John Tan",
bmi : 25.12,
age : 34,
nationality : true

(8 marks)

Question 1c

Modify the program in question 1a and 1b by converting the structure in question 1a to a C++ class. Implement a `main` function that performs the same tasks as required in question 1b using the C++ class you have just defined. Write your C++ code without changing the variable names described above. In addition, you may add your own variables if necessary but do not include other functions that are not needed here.

(12 marks)

MTD215 Copyright © 2022 Singapore University of Social Sciences (SUSS)

Question 2

Identify suitable coding modules in a simple program that implement polymorphism. In this question you are asked to apply operator overloading which is a form of polymorphism.

An instance of Soldier can be represented by height, weight and BMI. The specification of the three data members are as follows:

- Height – a float representing the height of the Soldier in m

- Weight – a float representing the weight of the Soldier in kg

- BMI – a float representing the bmi of the Soldier in bmi

(Note: bmi = weight / (height * height))

Addition

Two instances of Soldier can be added using the + operator. The weight and the height will be the average of the two instances. Consider the following example:

soldier1 = (70kg, 1.70m) BMI - 24.2215
soldier 2 = (80kg, 1.80m) BMI - 24.6914
soldier 3 = soldier 1 + soldier 2 = (75kg, 1.75m) BMI - 24.4898

Implement a C++ class to model Soldier as specified below:

Question 2a

Declare the data members height, weight and BMI.

(3 marks)

Question 2b

Write the constructor and destructor.
Note: BMI is to be calculated from weight and height

(6 marks)

Question 2c

Overload the + operator.

(6 marks)

Question 2d

Write the main() function to construct some Soldier instances and demonstrate the use of the + operator. Print out the BMI of the instances.

(5 marks)

MTD215 Copyright © 2022 Singapore University of Social Sciences (SUSS)

Question 3

Discuss the concept of arrays and inheritance by answering the following questions.

Assume the following definition and initialisation in the program:

string salesman[10] = {"Karter", " Tristin", " Rene", "Brock", "Antony", " Phillip", "Charles", "Adriel", "Gregory", "Rudy" }

int sales[10] = {905,1006,364,2300,1800,2000,1800,2400,364,2400 };

The salesman array stores the name of all the salesman in the company and sales array stores the sales made of the current month.

Sample of the output from the respective functions

```
The Total Sales is : $15339
The average Sales is : $1533.9
Salesperson(s) with the lowest Sales: Rene Gregory
Salesperson(s) with the highest Sales: Adriel Rudy
```

Question 3a

Create a C++ function and write C++ statements to carry out the following tasks and explain how you derive your solution.

- totalSales() -> Output the total sales for the current month.

(5 marks)

Question 3b

Create a C++ function and write C++ statements to carry out the following tasks and explain how you derive your solution.

- averageSales () -> Output the average sales for the current month.

(5 marks)

Question 3c

Create a C++ function and write C++ statements to carry out the following tasks and explain how you derive your solution.

- lowestSales() -> Output the the name(s) of the salesman with the lowest sales for the current month

(8 marks)

Question 3d

Create a C++ function and write C++ statements to carry out the following tasks and explain how you derive your solution.

- highestSales() -> Output the the name(s) of the salesman with the highest sales for the current month

(7 marks)

Question 4

Appraise the use of SDL for geometry rendering. SDL has primitive functions for drawing point, lines and rectangles. Write **TWO (2)** methods that utilise loops to efficiently draw multiple shapes as shown in both figures in question 4a and question 4b.

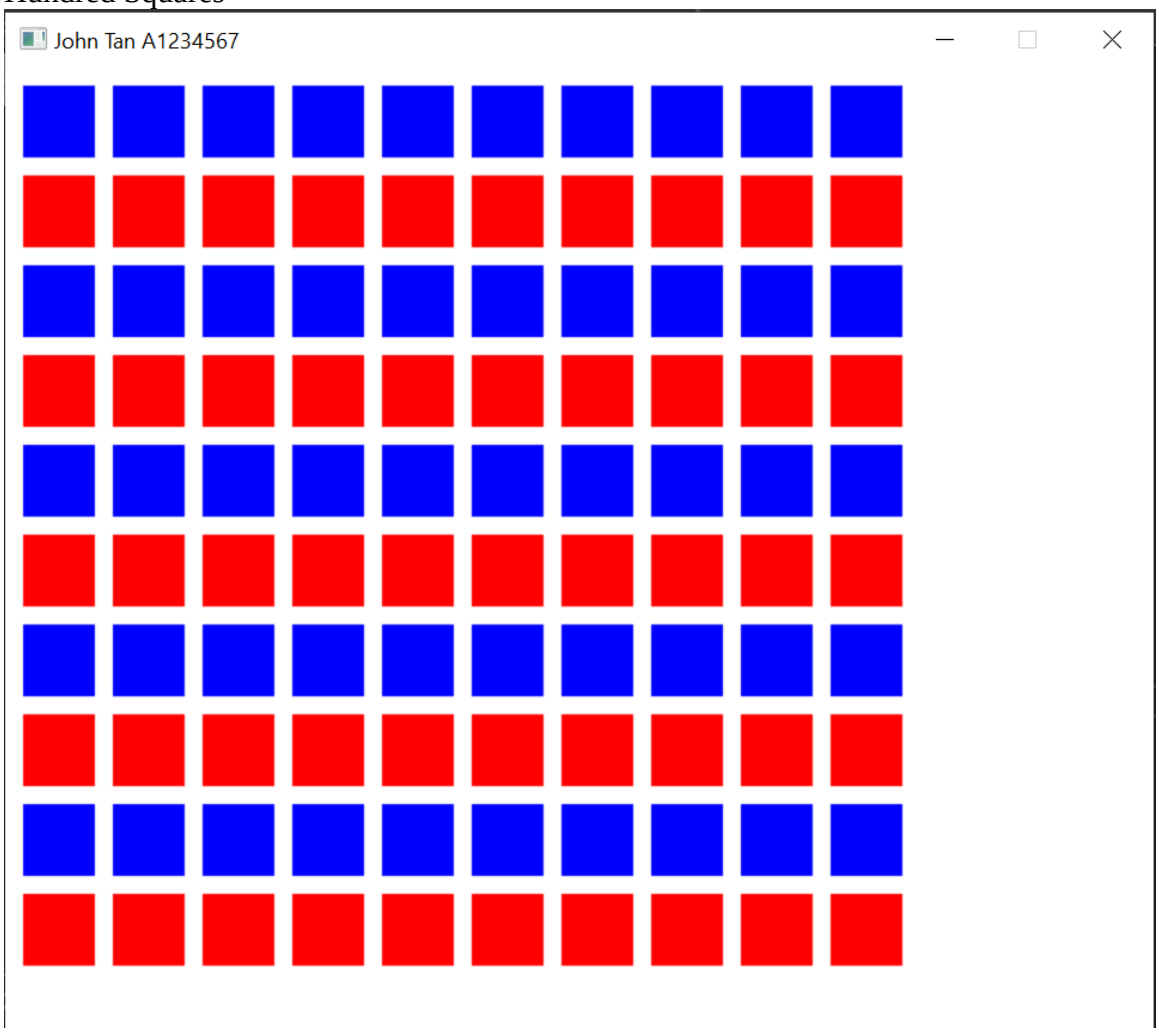
```
#define SDL_MAIN_HANDLED
#include <SDL2/SDL.h>

const int SCREEN_WIDTH = 640;
const int SCREEN_HEIGHT = 640;

SDL_Window* gWindow = NULL;
SDL_Renderer* gRenderer = NULL;


void drawMultipleRect() {
   //Your method implementation here
}

void drawMultipleTriange() {
  //Your method implementation here
}

int main(int argc, char* args[]) {
   SDL_Init(SDL_INIT_VIDEO);
   SDL_SetHint(SDL_HINT_RENDER_SCALE_QUALITY, "1");
   gWindow = SDL_CreateWindow("John Tan A1234567",
SDL_WINDOWPOS_UNDEFINED,
      SDL_WINDOWPOS_UNDEFINED, SCREEN_WIDTH, SCREEN_HEIGHT,
      SDL_WINDOW_SHOWN);
   gRenderer = SDL_CreateRenderer(gWindow, -1,
SDL_RENDERER_ACCELERATED);

   bool quit = false;
   SDL_Event e;
```

```
    while (!quit) {
        while (SDL_PollEvent(&e) != 0) {
            if (e.type == SDL_QUIT)
                quit = true;
        }

        SDL_SetRenderDrawColor(gRenderer, 0xFF, 0xFF, 0xFF, 0xFF);
        SDL_RenderClear(gRenderer);
        SDL_SetRenderDrawColor(gRenderer, 0x00, 0x00, 0x00, 0xFF);
      // drawMultipleRect();
      // drawMultipleTriange();


        SDL_RenderPresent(gRenderer);

    }

    SDL_DestroyRenderer(gRenderer);
    SDL_DestroyWindow(gWindow);
    gWindow = NULL;
    gRenderer = NULL;
    SDL_Quit();

    return 0;
}
```

Write the implementation for drawMultipleRect, drawMultipleTriangle function to produce each of following output respectively:

Question 4a

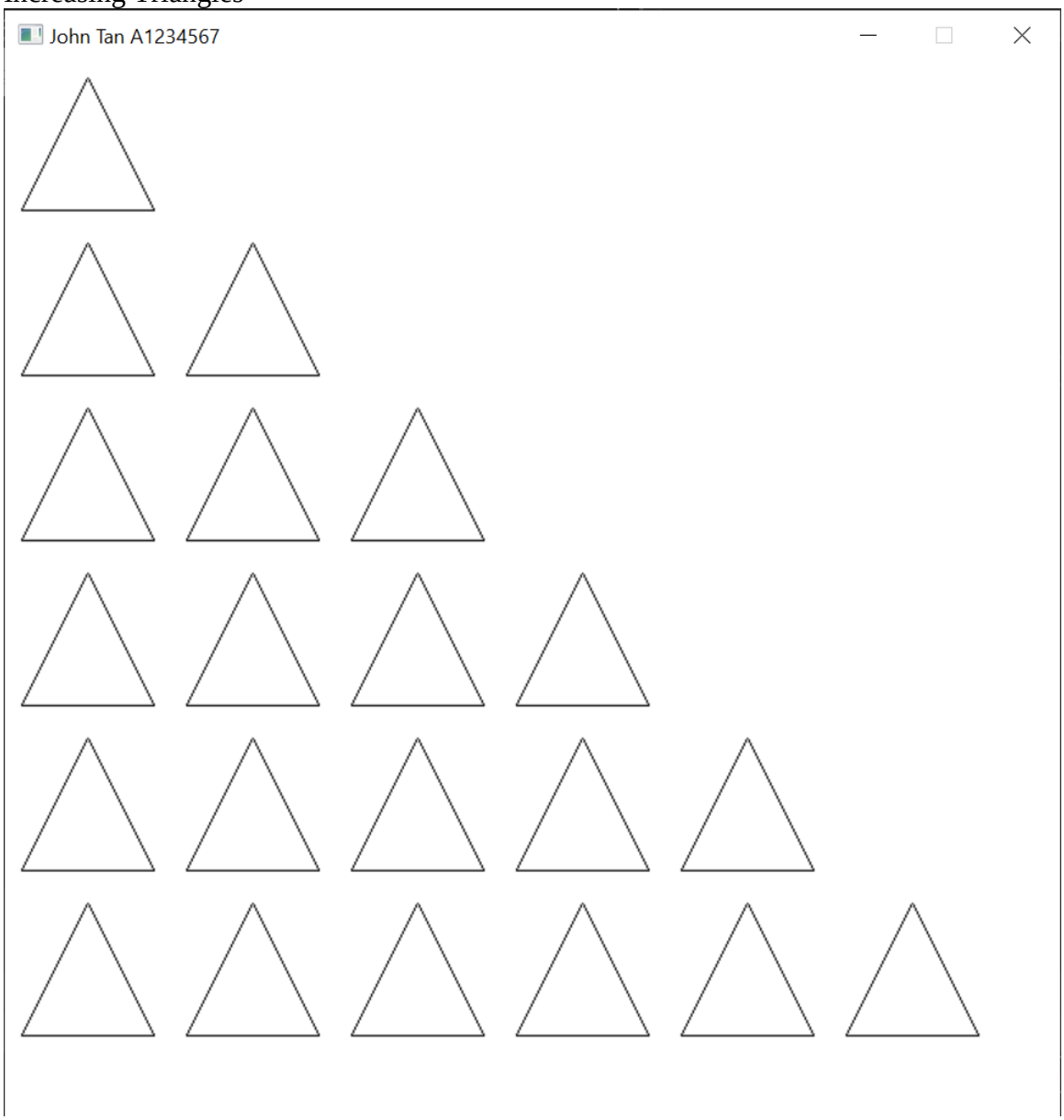
Hundred Squares



(15 marks)

Question 4b

Increasing Triangles



(15 marks)

----- END OF PAPER -----